FlashMAX III Solid-State Drive User Guide for Linux Systems

Document Number:	61600-00176-302
Document Version:	3.02
Software Version:	5.2.0
Revision Date:	22 December 2014

Warning: Printed copies of this document are considered current only on the date of print. Replacement and disposal of down-level versions is the responsibility of the document holder.

3rd Edition (Revision 3.02)

22 December 2014

The following paragraph does not apply to the United Kingdom or any country where such provisions are inconsistent with local law: HGST, INC., PROVIDES THIS PUBLICATION “AS IS” WITHOUT WARRANY OF ANY KIND, EITHER EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE. Some states do not allow disclaimer or express or implied warranties in certain transactions, therefore, this statement may not apply to you.

This publication could include technical inaccuracies or typographical errors. Changes are periodically made to the information herein; these changes will be incorporated in new editions of this publication. HGST may make improvements or changes in any products or programs described in this publication at any time.

It is possible that this publication may contain reference to, or information about, HGST products (machines and programs), programming, or services that are not announced in your country. Such references or information must not be construed to mean that HGST intends to announce such HGST products, programming, or services in your country.

Technical information about this product is available by contacting your local HGST representative or on the Internet at *http://www.hgst.com*.

HGST may have patents or pending patent applications covering the subject matter in this document. The furnishing of this document does not give you any license to these patents.

Conventions

The following icon and text conventions are used throughout this document to identify additional information of which the reader should be aware.

Conventions		Description
SHOCK HAZARD		***This icon indicates the danger of an electrical shock that may harm or otherwise prove fatal to the user***.
ESD		This icon indicates the possible presence of Electrostatic Discharge (ESD or "static electricity") that may harm the internal electronic components. The user is advised to handle the device only after discharging any possible electrostatic buildup that may be present.
CAUTION		This icon indicates the existence of a hazard that could result in equipment or property damage, or equipment failure if the safety instructions are not observed.
NOTE	√	This icon identifies information that relates to the safe operation of the equipment or related items.
Bold.	**Text**	Used to indicate **important technical notes**.
Bold Italic	***Text***	Used to indicate ***critical instructions***.
Light Blue Italic	*Text*	Used to indicate a *hyperlink* or *"jump"* to a related *section* or *subsection*.

Document Information

Product Classification:	FlashMAX III HH-HL SSD Cards
Date Last Updated:	December 22, 2014
Software Version:	5.2.0
Document Number:	61600-00176-302

Document Revision History

Version	Revised By	Revision Date	Description of Changes
5.2.0	HGST	12/22/2014	Initial release.

End-User License Agreement (EULA)

IMPORTANT: PLEASE READ THIS END USER LICENSE AGREEMENT CAREFULLY. DOWNLOADING, INSTALLING OR USING THE ACCOMPANYING SOFTWARE CONSTITUTES ACCEPTANCE OF THIS END USER LICENSE AGREEMENT.

HGST, INC. AND ITS AFFILIATES (COLLECTIVELY, "HGST") IS WILLING TO LICENSE THE SOFTWARE ONLY IF YOU ACCEPT ALL OF THE TERMS CONTAINED IN THIS END USER LICENSE AGREEMENT (THE "EULA").

BY DOWNLOADING, INSTALLING OR USING THE SOFTWARE OR OTHERWISE EXPRESSING YOUR AGREEMENT TO THE TERMS CONTAINED IN THIS END USER LICENSE AGREEMENT AND THE BROADER THE AGREEMENT (AS DEFINED BELOW), YOU INDIVIDUALLY AND ON BEHALF OF THE ENTITY THAT YOU REPRESENT (COLLECTIVELY, "LICENSEE") CONSENT TO BE BOUND BY THIS EULA AND THE AGREEMENT. IF YOU DO NOT OR CANNOT AGREE TO THE TERMS CONTAINED IN THIS AGREEMENT, THEN YOU WILL NOT HAVE ACCES TO THE SOFTWARE. CAPITALIZED TERMS USED, BUT NOT DEFINED, HEREIN SHALL HAVE THE MEANINGS GIVEN SUCH TERMS IN THE AGREEMENT.

1. Overview

No Sale: This EULA governs Licensee's use of and access to the Software (as defined below) for any Software first placed in use on or after the release date of this EULA (the "Release Date"). Licensee agrees and acknowledges that all Software is licensed and not sold, and that all use of "purchase" or "sell" in connection with any Software shall be deemed to mean "license." Any term in Licensee's purchase order or other document purporting to require the sale of Software instead of the license thereof is hereby rejected by HGST and is not part of the Agreement.

No Implied Licenses; Licensed Material: Nothing in this EULA shall be deemed to provide to Customer any right, title or interest in:

(i) any product sold by HGST or an authorized reseller of HGST (a "Product");

(ii) any instance of a program, module, feature, function, service, application, operation or capability of the software HGST intentionally delivers under this EULA ("Software"); or

(iii) any documentation related to any Software (the "Documentation"), other than the express license rights set forth in Section 4 hereof. All rights not expressly granted to Licensee are reserved by HGST and its licensors.

Delivery: Unless otherwise stated by HGST, Software will be delivered electronically.

Delivery is considered complete when HGST provides access to the Software download. For Users who purchase multiple licenses, delivery for all licenses is complete when HGST provides access to the first copy of the Software.

Fees: All fees paid are non-refundable.

Definitions: In this EULA, the following capitalized terms shall have the meaning set forth below: "**Agreement**" shall have the meaning set forth in Section 10.7.

"**Affiliate**" of a Party means such Party's parent corporation, an entity under the control of such Party's parent corporation at any tier or an entity controlled by such Party at any tier. For these purposes, "control" shall mean the power to direct or cause the direction of the management and policies of the entity, whether through the ownership of more than 50% of the outstanding voting interests in such entity or otherwise.

"**Commercial Software**" means software that HGST intends to be used by businesses.

"**Feature**" means an activity or class of activities that can be undertaken by the Software.

"**Party**" means either Licensee or HGST and "Parties" means both Licensee and HGST.

"**Software**" has the meaning set forth in Section 2.2.

"**Software Key**" means a code or key that is supplied by HGST allowing for the use of Commercial Software, which may provide capacity, device or other usage restraints consistent with the applicable order. For the avoidance of doubt, the term "Software" includes the applicable "Software Key".

"**Third Party**" means an individual or entity other than Licensee, Licensee's Affiliates, HGST and HGST's Affiliates.

“**Territory**” means the country in which Licensee received the Software, unless otherwise specified in a purchase order agreed to by HGST.

“**Update**” means an update to the Software that is supplied by HGST and that may incorporate:

(i) corrections of any substantial defects;
(ii) fixes of any minor bugs; and/or
(iii) at the sole discretion of HGST, minor enhancements to the Software; provided, however, that Updates shall not include Upgrades

Updates are generally identified by HGST by a change to the version number to the right of the first decimal point (e.g., Version 3.1 to 3.2).

“**Upgrade**” means a new release of the Software that incorporates substantial changes or additions that:

(i) provide additional value and utility;
(ii) may be priced and offered separately as optional additions to the Software; and/or
(iii) are not generally made available to HGST’s customers without a separate charge.

Upgrades are generally identified by HGST by a change to the version number to the left of the first decimal point (e.g., version 3.2 to 4.0).

“**User**” means an individual able to use or gain access to any Software functionality.

2. License Grant

Grant: Subject to the terms and conditions of this Agreement, HGST hereby grants to Licensee, and Licensee accepts, upon delivery of the Software, a nonexclusive, non-transferable license to install and use (subject to the limitations in Sections 4.5 and 4.6) each copy of the Software, in executable form only, supplied by HGST, and to use the accompanying Documentation, only for Licensee’s internal use and only as authorized in the Agreement.

Licensee may make a reasonable number of copies of the Software for inactive backup and archival purposes.

Trial Period: If Licensee receives the Software for a trial period, Licensee will be licensed to use the Software during the trial period. Subject to Section 7 below, if Licensee does not pay the applicable License fee and activate the Software, Licensee’s right to use the Software will end at the end of the trial period. The trial period will end either after the expiration of the trial period or when Licensee activates the Software, whichever occurs first.

Software Keys: HGST may provide Licensee with a Software Key to unlock the Software or certain Software Features agreed to by the Parties as part of the Agreement. Licensee’s license to the Software Key is limited to a license to use the Software Key to activate the Software only for the Features that HGST has agreed to activate. Any contractual limitations on the scope of Licensee’s license to Software shall apply regardless of whether a Software Key enforces such limitations.

Network Usage: Licensee understands and agrees that HGST may use Licensee’s internal network and Internet connection for the limited purpose of transmitting license-related data at the time of installation, registration, use or update of Software to a HGST-operated license server.

At such time, HGST may validate the license-related data in order to protect HGST against unlicensed or illegal use of the Software. HGST may, at its option, only permit activation of the Software upon exchange of license related data between Licensee’s computer and the HGST license server. Some Software may have license verification mechanisms that do not require the use of Licensee’s internal network and Internet connection, and HGST shall not use Licensee’s network or Internet connection for such Software.

3. Updates and Upgrades

Updates: Updates or Upgrades to the Software may be made available to Licensee pursuant to a separate agreement between HGST and Licensee. Any Update or Upgrade provided by HGST shall be licensed under the terms of the Software that is being updated by such Update or Upgrade, as the case may be.

No Obligation: Nothing in this EULA requires HGST to provide Updates or Upgrades to Licensee or Licensee to accept such Updates or Upgrades. The provision of any Updates or Upgrades shall be governed by a separate agreement between HGST and Licensee, or by a purchase order issued by Licensee and accepted by HGST, in HGST’s sole discretion, and may require the payment of additional fees.

No Right to Sublicense or Assign: Except to the extent otherwise required by applicable law or expressly provided for assignment generally in the Agreement, including without limitation in this EULA, no license provided in this Section

4 is sub-licensable, transferable or assignable by Licensee, including by operation of law, change of control, merger, purchase or otherwise, without the prior written consent of HGST in each instance. Other than as expressly permitted by the foregoing, any attempted sublicense, transfer or assignment by Licensee shall be null and void.

License Prohibitions: Notwithstanding anything to the contrary in this EULA, Licensee shall not, alone, through a User, an Affiliate or a Third Party (or allow a User, an Affiliate or a Third Party to):

- Sell, transfer, license, lease pledge, loan or otherwise make available to Third Parties any Software;
- Modify any Software;
- Reverse compile, reverse assemble, reverse engineer or otherwise translate all or any portion of any Software;
- Pledge, rent, lease, share, distribute, sell or create derivative works of any Software;
- Use any Software on a time sharing, service bureau, application service provider (ASP), rental or other similar basis;
- Make copies of any Software, except as provided for in the license grant above;
- Remove, alter or deface (or attempt any of the foregoing) proprietary notices, labels or marks in any Software;
- Distribute any copy of any Software to any Third Party, including without limitation selling any Product with Software in a second-hand market;
- Disclose the results of testing or benchmarking of any Software to any Third Party without the prior written consent of HGST;
- Use any Update or Upgrade beyond those to which Licensee is entitled or with any Software to which Licensee does not have a valid, current license;
- Circumvent or disable HGST's copyright protection mechanisms or license management mechanisms;
- Use the Software in violation of any applicable law or to support any illegal activity; or
- Attempt any of the foregoing.

HGST expressly reserves the right to seek all available legal and equitable remedies to prevent any of the foregoing and to recover any lost profits, damages or costs resulting from any of the foregoing.

Maximum Usage: Licensee shall not use Software in excess of the amount licensed to Licensee, as specified in the Agreement, which may be on a per user, per concurrent active device, capacity or other metric as established under the Agreement.

Legal Exception: Licensee agrees that, to the extent that any applicable laws (including without limitation national laws implementing EC Directive 91/250 on the Legal Protection of Computer Programs) give Licensee the right to reverse engineer any Software to make it interoperable without HGST's consent, before Licensee exercises any such rights, Licensee shall notify HGST of such desire and, no later than sixty (60) days following receipt of such request, HGST may decide either:

(a) to perform the work to achieve such interoperability and charge its then-standard rates for such work to Licensee; or

(b) to permit Licensee to reverse engineer parts of the Software only to the extent necessary to achieve such interoperability.

Only if and after HGST, at its sole discretion, partly or completely denies Licensee's request, shall Licensee exercise its statutory rights.

Third Party Software: Portions of the Software may be owned by one or more Third Parties and licensed to HGST. HGST and Licensee intend and agree that Software owned by Third Parties and supplied hereunder is being sublicensed to Licensee, that such Third Parties retain ownership of and title to such Software, network and that such Third Parties may directly enforce Licensee's obligations hereunder to protect their respective interests in such Software. The warranty and indemnification provisions set forth herein shall not apply to Software owned by Third Parties and supplied hereunder. If portions of the Software are owned by one or more Third Parties, the licenses/terms applicable to such Third Party licenses are set forth in Exhibit A.

UNLESS OTHERWISE AGEED TO BY THE THIRD PARTY, TO THE MAXIMUM EXTENT PERMITTED BY LAW, THIRD PARTY SOFTWARE COMPONENTS AND DOCUMENTATION ARE PROVIDED "AS IS". NEITHER HGST NOR THE THIRD PARTY PROVIDER MAKES ANY WARRANTY RELATED TO THE THIRD PARTY SOFTWARE COMPONENT, STATUTORY, EXPRESS OR IMPLIED, AND EACH SPECIFICALLY DISCLAIMS THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT WITH RESPECT TO THE SOFTWARE AND THE DOCUMENTATION.

4. Product Specific Terms

Commercial Software Audit: Licensee hereby grants to HGST, itself or using an independent third party auditor, the right to audit Licensee's books, records and systems during normal business hours upon seven (7) days' prior written notice, no more often than once per annum regarding the usage of Commercial Software under the Agreement. In the event such audit identifies material non-compliance with this EULA or the Agreement between Licensee and HGST, Licensee shall pay HGST's costs of such audit.

Confidentiality: Subject to any other confidentiality restrictions set forth in the Agreement, Licensee agrees the Software and Documentation are the confidential information of HGST. Licensee shall maintain the Software and Documentation in confidence, using the same degree of care it uses for its own confidential information of a similar nature, which shall be no less than reasonable and customary efforts in the software industry.

Software Warranty: TO THE MAXIMUM EXTENT PERMITTED BY LAW, EXCEPT AS PROVIDED FOR IN THE AGREEMENT, THE SOFTWARE AND DOCUMENTATION ARE PROVIDED "AS IS". HGST MAKES NO OTHER WARRANTIES, STATUTORY, EXPRESS OR IMPLIED, AND SPECIFICALLY DISCLAIMS THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT WITH RESPECT TO THE SOFTWARE AND THE DOCUMENTATION. IN ADDITION, HGST DOES NOT WARRANT THAT THE SOFTWARE'S OPERATION WILL BE UNINTERRUPTED OR ERROR-FREE, OR THAT ALL DEFECTS WILL BE CORRECTED.

5. Disabling Code

Disabling Code: Software may be provided to Licensee with disabling code that allows HGST to disable such Software. Any Updates or Upgrades to the Software may include disabling code. HGST agrees not to invoke such disabling code except as provided for in Section 7.2, without Licensee's prior consent, which may be given by telephone or email.

Invocation of Disabling Code: In addition to the invocation of disabling code when HGST has received Licensee's consent described in Section 7.1, HGST may, at its option, invoke disabling code in HGST's Software without receiving Licensee's consent

(i) if in HGST's sole, reasonable discretion, HGST believes that such Software has been, is being or will be used in violation of laws;
(ii) if HGST is required to do so, because of a court or regulatory order;
(iii) if Licensee has not paid the applicable License Fee by the expiration of the Software trial period; or
(iv) if Licensee has used the Software other than as authorized by Licensee's license

HGST shall have no liability to Licensee for any good faith invocation of any such disabling code.

6. Exclusive Remedies and Limitation of Liability

Definitions: For purposes of the exclusive remedies and limitations of liability set forth in this Section 8, HGST shall be deemed to include its directors, officers, employees, agents, representatives, shareholders, subcontractors, licensors and suppliers; and "damages" shall be deemed to refer collectively to all injury, damage, loss or expense incurred.

Maximum Liability: Except as may otherwise be provided in the Agreement, HGST's maximum aggregate liability arising from or relating to this EULA or the use of the Software and Products shall not to exceed the total amount paid by Licensee to HGST for the applicable Software during the twelve (12) month period that immediately preceded the event that gave rise to the applicable claim.

Limitation of Liability: NOTWITHSTANDING ANY OTHER PROVISION OF THIS EULA BUT SUBJECT IN ALL RESPECTS TO THE AGREEMENT, TO THE MAXIMUM EXTENT PERMITTED BY LAW, HGST SHALL NOT BE LIABLE FOR INCIDENTAL, SPECIAL, EXEMPLARY, CONSEQUENTIAL OR OTHER INDIRECT DAMAGES, INCLUDING BUT NOT LIMITED TO THE COST OF PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES, LOSS OF DATA, LOSS OR INTERRUPTION OF USE, LOST PROFITS, SAVINGS OR REVENUES OF ANY KIND, WHETHER OR NOT HGST HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THIS PROVISION SHALL APPLY EVEN IN THE EVENT OF THE FAILURE OF AN EXCLUSIVE REMEDY.

7. Term and Termination

Term: The term of this EULA is while any Software is under Licensee's control or possession, unless terminated earlier (the "License Term").

Termination: HGST shall have the right to terminate this EULA:

(a) upon thirty (30) days' prior written notice to Licensee if Licensee has not cured any material breach of this EULA by the end of such thirty (30) day notice period;

(b) if Licensee has not paid an invoice to HGST regarding such Software sixty (60) days after such invoice is due, at any time thereafter upon written notice to Licensee;

(c) if the Software is found to infringe or misappropriate any Third Party's intellectual property or proprietary rights, or becomes subject to an injunction prohibiting its use, or HGST reasonably believes either of the foregoing conditions is likely to occur; or

(d) as otherwise provided for in the Agreement. Upon termination of this EULA for any reason,

(i) Licensee shall be responsible for payment for all purchase orders delivered to Licensee by HGST before the effective date of termination; and

(ii) Licensee shall destroy all copies of the Software under Licensee's control or possession, and provide HGST with reasonably requested evidence of the destruction of copies of the Software.

Survival: The provisions of Sections 1, 2, 4.5, 4.6, 4.8, 4.9, 5, 6, 7, 8, 9.3, and 10 of this EULA shall survive any termination in accordance with their terms, in addition to any accrued obligations.

8. Miscellaneous

Choice of Law; Jurisdiction: The Agreement shall be governed by the laws of the State of California, excluding:

(a) its conflicts of law rules that would result in the application of the law of any other jurisdiction; and

(b) the United Nations Convention for the International Sale of Goods.

The federal and state courts located in San Jose, California shall have exclusive jurisdiction with respect to any dispute arising under this Agreement.

Assignment: Neither Party may assign its rights and obligations hereunder without the prior written consent of the other Party.

Notwithstanding the foregoing, either Party may assign this EULA to any Affiliate of the other or to an acquirer (by purchase, merger or otherwise) of all or substantially all of such Party's business or assets relating to this EULA, provided that:

(i) the assignee agrees in writing to be bound by the terms and conditions of this EULA

(ii) neither the assignor nor assignee are in default hereunder. Any attempted assignment other than as permitted shall be null and void.

Non-Waiver: No course of dealing or failure of either party to strictly enforce any term, right or condition of this Agreement shall be construed as a waiver of such term, right or condition. In the event of any breach or threatened breach of Section 3, Licensee agrees that HGST will suffer irreparable damage for which HGST will have no adequate remedy at law. Accordingly, in addition to any other remedy, HGST shall be entitled to injunctive and other equitable remedies to prevent or restrain such breach or threatened breach, without the necessity of proving actual damages or posting any bond.

Export Regulations: Licensee understands that HGST is subject to regulation by the U.S. government and its agencies, which prohibit export or diversion of certain technical products and information to certain countries and individuals. Licensee warrants to HGST that Licensee will comply in all respects with all export and re-export restrictions applicable to the technology and documentation provided hereunder.

Restricted Party: Licensee represents to HGST that Licensee is not a Restricted Party, which shall be deemed to include any person or entity:

(1) located in or a national of Cuba, Iran, Libya, North Korea, Sudan, Syria, or any other countries that may, from time to time, become subject to U.S. export controls for anti-terrorism reasons or with which U.S. persons are generally prohibited from engaging in financial transactions;

(2) on the U.S. Department of Commerce Denied Person's List, Entity List, or Unverified List; U.S. Department of the Treasury list of Specially Designated Nationals and Blocked Persons; or U.S. Department of State List of Debarred Parties;

(3) engaged in activities involving nuclear materials or weapons, missile or rocket technologies, or proliferation of chemical or biological weapons; or

(4) affiliated with or a part of any non-U.S. military organization. Licensee shall not sell, resell, loan, disclose, or otherwise transfer any Software, Product or Documentation to any Restricted Person without prior, express written authorization from HGST and the appropriate U.S. Government agency. If Licensee becomes a Restricted Person during the Term, this EULA shall automatically terminate and Licensee shall immediately cease all use of the Software.

9. United States Government Restricted Rights

The Software and Documentation is provided with restricted rights and qualifies as "commercial items" consisting of "commercial computer software" and "computer software documentation" as such terms are defined and used at FAR (48 C.F.R.) 2.101 and FAR 12.212. Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(f)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.227-7013 or subparagraphs (c) (1) and (2) of the Commercial Computer Software-Restricted Rights at 48 C.F.R. S:52.227-19, as applicable.

For clarity, this Section 9.6 shall also be construed to apply to the state or local government counterparts, or the foreign counterpart specified as the Territory, as applicable.

10. Order of Precedence; Entire Agreement

To the extent of any conflict among this EULA, any additional terms in an agreement signed by Licensee and HGST, any HGST "click-accept" agreement, any terms on a purchase order and HGST's terms and conditions of sale, the order of precedence shall be:

(a) an agreement signed by Licensee and HGST; (b) this EULA;

(b) a "click-accept" agreement, including any agreement regarding the Features to be included as part of the Software;

(c) HGST's terms and conditions of sale; and

(d) HGST's purchase order acceptance (collectively,

(a)-(e), after applying the order of precedence, the "Agreement").

The terms and conditions contained in the Agreement supersede all prior oral or written understandings between the Parties and shall constitute the entire agreement between the Parties with respect to the subject matter of the Agreement. This Agreement shall not be modified or amended except by a writing signed by Licensee and HGST.

Construction; Severability: The headings used in this Agreement are for reference purposes only and will not be deemed to limit, expand or in any way affect the interpretation of any term or provision hereof. If any provision or part hereof shall be held to be invalid or unenforceable for any reason, then the meaning of such provision or part hereof shall be construed so as to render it enforceable to the extent feasible. If no feasible interpretation would save such provision or part hereof, it shall be severed here from, but without in any way affecting the remainder of such provision or any other provision contained herein, all of which shall continue in full force and effect unless such severance effects such a material change as to render the Agreement unreasonable.

Table of Contents

List of Figures

List of Tables

1. FlashMAX III Overview

1.1 Overview

This section introduces the basic, advanced and administrator level technical specifications of the FlashMAX III SSD (Solid-State Drive) Card.

1.2 Audience

This manual is intended for system engineers, system designers, or Information Technology (IT) professionals employed by an Original Equipment Manufacturer (OEM). This document was therefore written for a technically advanced audience; it is not intended for end-users that will eventually purchase the commercially available product. The ***user***, as referred to throughout the manual, is primarily concerned with industrial, commercial, or military computing applications.

1.3 FlashMAX III Series

The FlashMAX III® Series of solid-state storage devices that combine the industry standard PCI Express bus interface with vFAS, a hardware and software architecture designed to deliver the highest sustained application performance over its lifetime and enterprise-class reliability. FlashMAX III enables enterprise vendors to tackle performance-intensive applications such as databases, business analytics, virtualization, and high-performance computing applications.

1.4 Design Description

FlashMAX III is a Half-Height, Half-Length (HHHL) PCIe 3.0 x8 Lane Application Accelerator (AA). The device is available in capacities up to 4.8TB, making it the largest available solid-state AA in this form factor.

The devices are integrated with a new software suite, FlashMAX Connect, which is server-side software that enables additional features like high availability, shared storage management, and data caching.

1.5 Usable Capacity and Operational Modes

The actual usable capacity is dependent on the operational mode. The user can choose whether the card operates in *Maximum Capacity Mode* or *Maximum Performance Mode*. Table 1 lists the differences in the usable capacity according to the operational mode.

Table 1: Usable Capacity and Operational Mode

Maximum Capacity Mode	Maximum Performance Mode
1,111GB (1.1TB)	924GB (0.93TB)
2,222GB (2.2TB)	1,848GB (1.84TB)

1.6 Physical Characteristics

Figure 1 shows the front side view of a FlashMAX III card having a capacity of 2.2TB.

√ The physical appearance varies from card to card, depending on the capacity, e.g., a higher capacity card might consist of two PCBs, rather than just one.

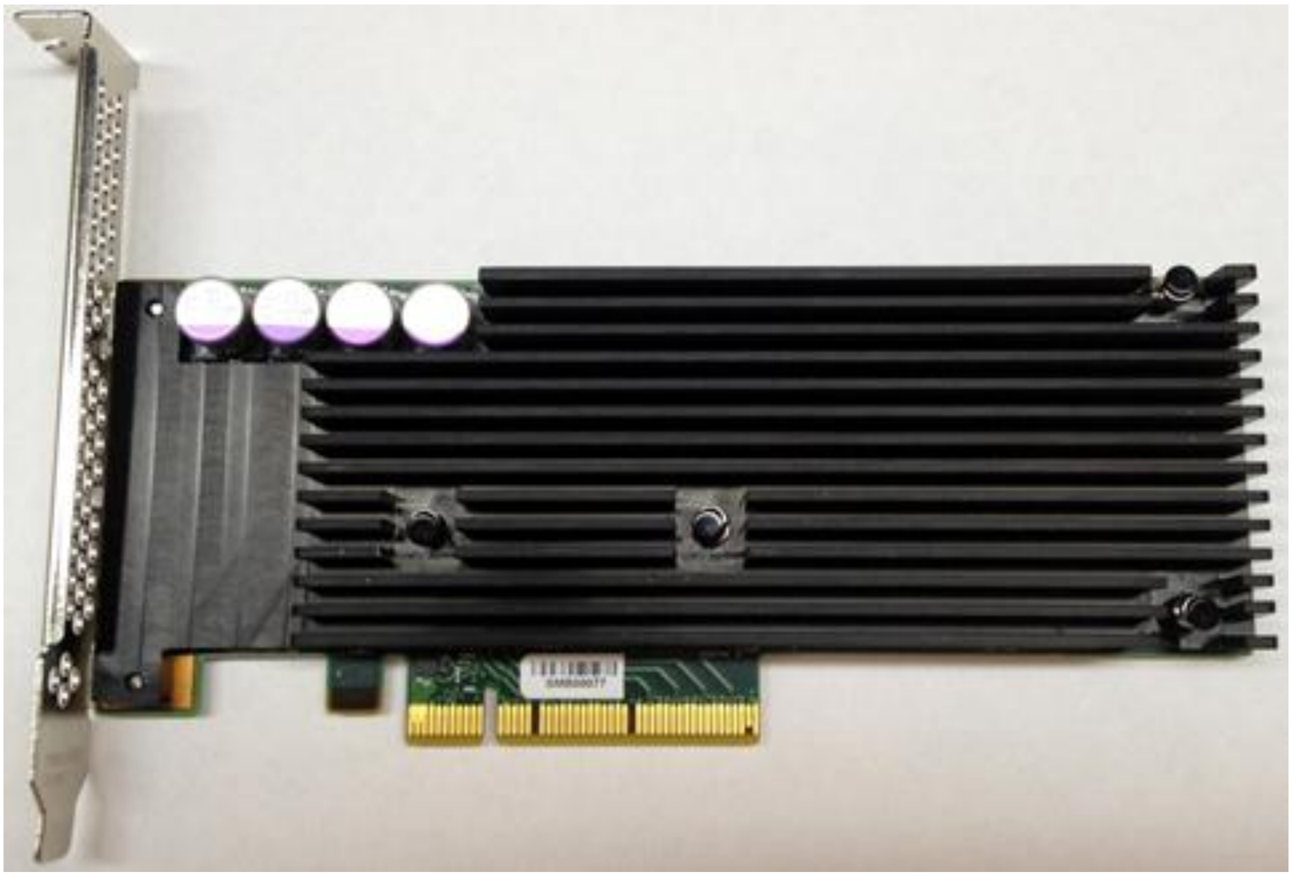

Figure 1: FlashMAX III Card - Front Side View

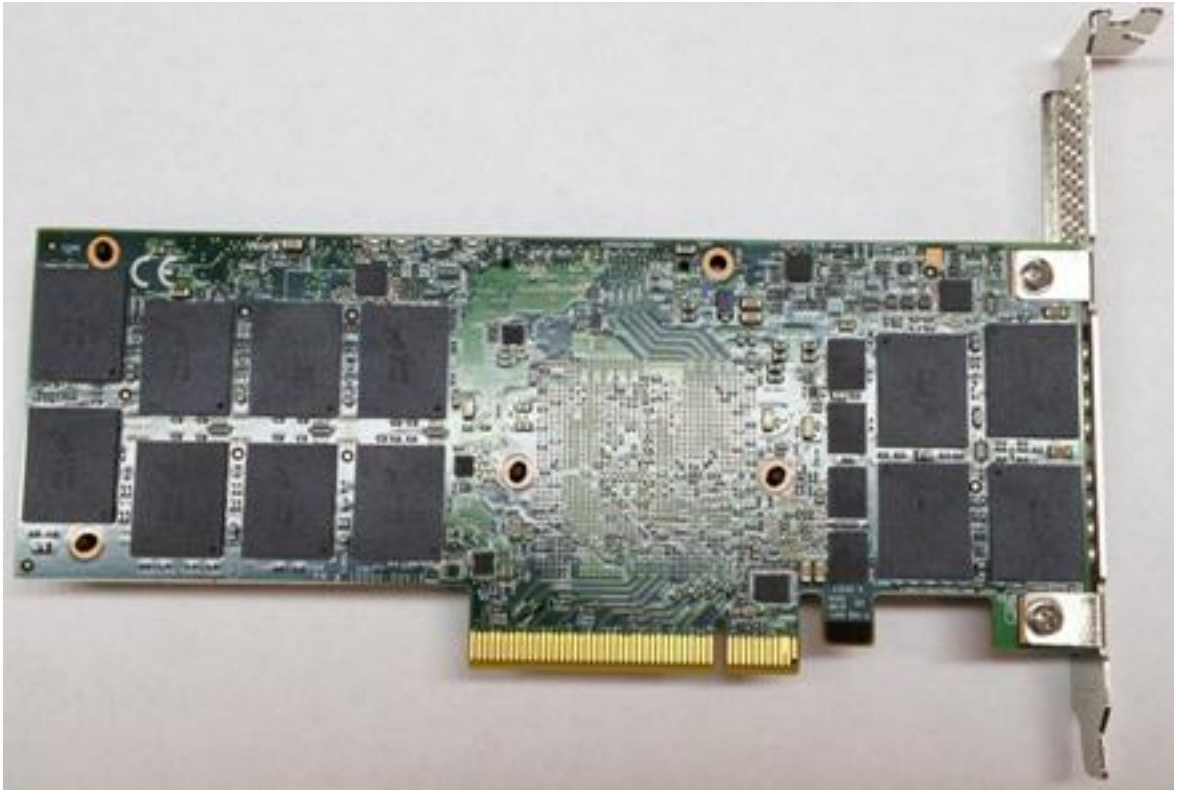

Figure 2: FlashMAX III Card - Reverse Side View

Figure 3: FlashMAX III Card - Bottom Side View

1.7 System Requirements

Table 2 lists the system hardware requirements needed to provide optimal performance.

√ The RAM requirements listed in the following table are only applicable to the FlashMAX devices. The OS and applications will require additional RAM.

√ All FlashMAX models are compatible with all multi-core (Server) Intel® or AMD™ processors, such as the Intel Xeon® or AMD Opteron™.

Table 2: Standard System Requirements

Model	Capacity	PCIe Type	Air Flow	RAM
FlashMAX III	1.1TB	Gen2 x8 Electrical	150LFM @ 45C Inlet	3GB for 1.1TB
FlashMAX III	2.2TB	Gen2 x8 Electrical	200LFM @ 45C Inlet	4GB

2. Installation

2.1 Overview

This section outlines the procedure for installing the FlashMAX III card.

2.2 Card Bracket

Make sure the appropriate bracket is attached to the card before beginning the installation:

Table 3: Bracket Types and Descriptions

Bracket Type	Description
Full-Height Bracket (Standard)	A full-height (dual-unit) bracket is included in the standard shipment.
Half-Height Bracket (Spare)	A spare half-height (single unit) bracket is provided in the box.
Half-Height Bracket (Alternate)	The half-height (single unit) bracket can be substituted for the full-height bracket.
Replacing the Bracket	*See Changing the Bracket.*

2.3 Changing the Bracket

Figure 4 indicates the location of the bracket screws. To replace the bracket, remove the screws as indicated by the ***red circles*** in the figure.

Do not remove the heat sinks while removing the bracket. Removal of the heat sinks may cause damage to the device and void the warranty.

√ *An m2.5 screw driver and m25 nut driver are needed to remove the mounting bracket hardware.*

√ *The screws have a torque specification of 4lb-in or 0.45N-m.*

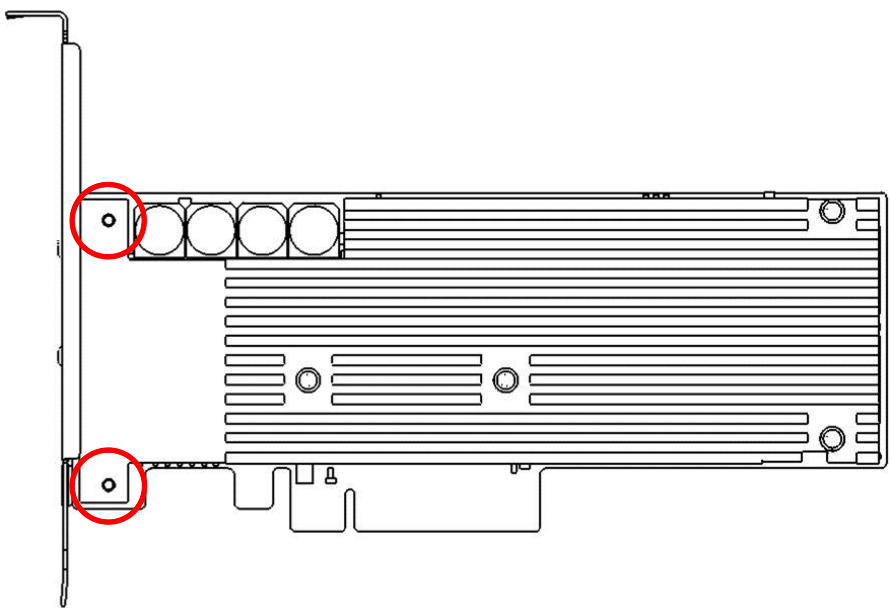

Figure 4: Bracket Screw Locations

2.4 Safety Instructions

This section addresses the issues in regards to electrical safety during the hardware installation.

There is a risk of electrocution! Use extreme caution when handling the solid-state drive and while connecting it to a power source. Make sure to read and thoroughly understand this section before attempting to install the drive.

Observe all applicable electrical safety rules while installing the solid-state drive.

Electrostatic Discharge or ESD can seriously damage the electronics of the host system, enclosure, and solid-state drive. It is very important that the user discharge any "static electricity" before starting the installation procedure. The user can touch an unpainted, grounded metallic surface to discharge any static charges that may be present on the body or clothing. As an alternative, the user can also wear an ESD protective wrist strap. The user can minimize the possibility of damage due to ESD by avoiding physical contact with the electronic components and interface connector.

To install the card the server:

1. ***Power off the system.***
2. ***Disconnect the AC power source.***
3. If necessary, remove the access cover.
4. Check the slot power requirements. *See Selecting a PCI Slot.*
5. ***Make sure the card is in the correct orientation before inserting it into the drive bay or into the PCIe bus slot.***
6. ***The connector on the card is keyed to ensure that the signal and power connections to the device are correctly oriented.***
7. ***Do not force the card into the drive bay, backplane connector, or PCIe bus slot.*** Doing so may damage the device, bus slot, enclosure, or system board beyond repair.
8. ***Do not over tighten the bracket screw, but apply sufficient torque to ensure the device is secure.*** Secure the device in the drive bay or PCIe bus slot using the recommended machine screw.
9. If necessary, replace the access cover.
10. Power on the system.

2.5 Selecting a PCIe Slot

The system may have multiple types of PCIe slots. The card is designed to install in most PCIe slots; however, the choosing the wrong slot may have a negative impact on the overall functioning of the card.

√ *Make sure to choose a slot that conforms to the air flow requirements and ventilation.*

√ *Make sure that while installing the card that any other hardware components do not block the air flow around it.*

Default PCIe:	The card can be installed in a PCIe Gen3 x8 Lane or PCIe Gen3 x16 Lane slot.
Alternative PCIe:	If there are no PCIe Gen3 x8/x16 slots, then slots that have x8 physical connectors using x4 electrical connectors can be used.
Repercussions:	Using a PCIe x4 Lane slot may result in up to a 50% reduction in performance.
Technical Fixes:	*See Additional Information.*

2.6 Additional Information

- A PCIe slot having x8 physical connections might be using only x4 electrical connectors; the slot will behave like an x4 slot.
- Any such PCIe slots having a mixed combination of physical and electrical connectors affects overall system functioning and performance.
- It is advised to reference the system documentation for details.

3. Configuring System BIOS Settings

3.1 Overview

Once the card installation is complete the system must be restarted and BIOS modified. As the system restarts and begins the bootstrapping process, it will display a set of *Function Keys* ([F1], [F2], [F3], etc.) on the screen that are used for accessing the system BIOS. It is recommended that the user review the system documentation beforehand to learn which function key allows for system BIOS access.

√ Not all systems use the same assigned function keys to enter BIOS setup.

√ Some AMIBIOS systems use the Delete or [Del] key to enter BIOS setup.

3.2 Additional Information

- Each BIOS setup utility has a unique UI screen, and some of the procedures as described here may not accurately reflect the BIOS setup on the system.
- Please go to the *HGST Support Portal* for specific BIOS screenshots and installation details.

3.3 Changing the Fan Speed Settings

3.3.1 General Instructions

Proper ventilation and cooling is of critical importance to the effective functioning of any electrical or electronic equipment or system. Heat sinks and system fans control the operating temperature environment. In regards to computer systems and servers, the performance and functioning of the system fans can be configured through the BIOS settings.

√ *If the BIOS settings allow for the speed configuration of the system fans, then set the fan speeds to **maximum**.*

3.3.2 Hewlett-Packard Systems

For Hewlett-Packard (HP) systems, the setup is usually:

BIOS -> Advanced Options ->, Thermal Configuration -> Increased/Maximum Cooling

3.3.3 Dell Systems

For Dell systems, the setup is usually:

System Settings -> IDRAC Settings ->, Thermal, with a set of options of:

- Maximum Performance
- Fan Speed Offset
- High Fan Speed Offset

3.4 Changing the Processor Settings

The following settings are usually found under the “Advanced: Processor/Chipset Control” BIOS menus.

Disable the following optional settings:

- Processor Performance States / C States / C-State Tech / EIST
- C1E States / C1 Enhanced States
- Active State Power Management

Enable the following optional settings:

- Hyperthreading / Logical Processors
- Turbo Mode

√ *If there are Performance Profiles (System Power Modes) available, then the user should use the “Maximum Performance” setting.*

3.5 Third-Party Information

Issue:	Intel has identified specific issues with the Xeon E5-2600 series processors.
Impact:	Severe drops of PCIe and FlashMAX III bandwidth may occur when running workloads that have low CPU consumption.
Technical Fix:	To avoid this issue, the C2 and C1E processor power states must be disabled.
More Information:	*See the Erratum in the Intel® Xeon® Processor Family Specification Update, September 2014, Revision 017.*

4. FlashMAX II Compatibility

4.1 Overview

This section explains the steps needed to install and configure a FlashMAX III card when one or more FlashMAX II (V2) card(s) is/are pre-installed in the system. This section assumes that the system has been configured with pre-installed FlashMAX II cards.

4.2 Supported Functionality

Functionality has been introduced to support compatibility between FlashMAX II and FlashMAX III drivers when both products are installed in the system. The driver and utilities have been modified to support the following requirements:

- Enabling co-existence of multiple FlashMAX II and FlashMAX III drivers in any given system.
- The base of all the device names exposed is Gateway ID. The Gateway ID will be distributed to both FlashMAX II and FlashMAX III via bus drivers.
- The enumeration of the devices for FlashMAX II and FlashMAX III cards should be consistent across the reboot.

√ This feature allows multiple FlashMAX II and FlashMAX III cards in any combination on a given system.

√ This feature holds good for existing systems where one or more FlashMAX II card(s) already exist and new FlashMAX III card(s) will be added.

4.3 Additional Information

- The feature has been developed with the approach that a common driver package is suitable to support a co-existence scenario.
- In the scenario where one or more FlashMAX II card(s) exist in the system, the same requirement for re-installed using the new driver package applies.

4.4 Enabling Co-Existence Considerations

The user should make the following considerations when enabling co-existence.

The co-existence support for FlashMAX III cards is only limited until its FlashMAX II version.

V1 cards currently installed in the system will not support co-existence with FlashMAX III cards; hence, the V1 cards would need to be removed from the system.

4.5 Managing the Compatibility

If compatibility must be enabled, the existing FlashMAX II cards must be first uninstalled. The required number of FlashMAX II and FlashMAX III can be reinstalled using the latest driver set and utility package.

4.5.1 Uninstalling FlashMAX II Cards

To uninstall and remove the FlashMAX II cards, follow the steps as described in *Section 6.3*.

4.5.2 Installing FlashMAX II and FlashMAX III Cards

To install FlashMAX II and FlashMAX III cards, follow the steps as described in *Section 7.2*.

5. Operating System Configuration

5.1 Overview

This section explains how to change the OS shell and kernel settings for a FlashMAX III card. The user should have basic to advanced knowledge of OS shell and kernel settings.

5.2 Ubuntu

While the device driver for FlashMAX III supports both "`-server`" and "`-generic`" kernels, it is recommended that the "`-generic`" version be used to achieve better performance.

5.2.1 Third-Party Information

Issue:	Any Ubuntu Linux distribution.
Impact:	Use the "`-generic`" version of the kernel to achieve optimal performance.
Technical Fix:	Modify the default kernel installation to "`-generic`".
Benefit:	The "`-generic`" version manages highly threaded IO drivers more efficiently than the "`-server`" version.
More Information:	Use the `apt-get` command to change the installed version.

5.2.2 How To

To determine which kernel is presently installed, use the command:

`uname -r` and check whether the kernel revision ends in `-generic` or `-server`.

To install the generic version of the kernel, use the command:

```
apt-get install linux-generic
```

5.2.3 Command Prompt Instructions

The `apt-get` command is a command-line tool that interoperates with the Ubuntu Advanced Packaging Tool (APT). It performs functions such as installation of new software packages, upgrade of existing software packages, updating of the package list index, and is capable of upgrading the entire Ubuntu system.

`# apt-get install {package}`	Install new package.
`# apt-get remove {package}`	Remove/delete existing package.
`# apt-get --purge remove {package)`	Remove/Delete installation, including the config file.
`# apt-get update`	Resynchronize the package index files.
`# apt-get upgrade`	Upgrade the Debian Linux system.
`# apt-get update`	Upgrade to Debian distribution.
`# apt-get dist-upgrade`	Upgrade to Debian distribution.

5.3 SUSE Linux

While HGST has already tested the FlashMAX III device driver on the SUSE Linux, Novell has not yet certified the driver; the user must enable the unsupported modules to allow for driver loading.

5.3.1 Third-Party Information

Issue:	SUSE Linux installation on Novell.
Impact:	SUSE will not allow such modules to be loaded.
Technical Fix:	The user must modify the boot file to allow the FlashMAX III device driver to load.
How to Fix:	*See the Command Prompt Instructions.*

5.3.2 How To

1. Edit the file `/etc/modprobe.d/unsupported-modules`.
2. Set the value `allow_unsupported_modules` to `1` as shown below.

5.3.3 Command Prompt Instructions

`vi /etc/modprobe.d/unsupported-modules`

`# allow_unsupported_modules 1` enables loading of unsupported by the `modprobe` setting.

`# allow_unsupported_modules 0` will disable it.

This can be overridden using the `--allow-unsupported-# modules` command line switch

`# allow_unsupported_modules 1`

5.4 Disabling CPU Throttling

The user should disable the dynamic CPU frequency scaling to obtain maximum performance. This is especially important when measuring I/O latencies.

To disable the CPU speed daemon in Red Hat Linux or CentOS:

```
service cpuspeed stop
chkconfig cpuspeed off
```

6. Software Utility Installation

6.1 Overview

This section explains how to install and configure the FlashMAX III utility software and drivers.

6.2 Preparation

√ The user must have a login account to download any software, driver, or utilities from the *HGST Support Portal*.

√ The user can request an account if there is not already and existing account.

√ The *HGST Support Portal* is a service provided free of charge.

The following steps are recommended before starting the installation of the utility software and drivers:

1. Visit the *HGST Support Portal* to ensure that the latest compatible driver(s) and utilities are available.
2. Visit the *HGST Support Portal* if necessary download the latest driver(s) and utility packages.
3. The user must login as `root` or `sudo bash` as applicable to execute all commands.

6.3 Option ROM

The Option ROM is enabled when the firmware is upgraded. The Option ROM is only needed if the user intends to boot the operating system from the card.

It is recommended to disable the Option ROM unless the user intends to boot from the card. The procedure is outlined in Appendix 1 of the UEFI Boot Guide.

6.4 Verify the Device Installation

The user must verify that the device has been detected and initialized by the PCIe subsystem after power-on. The `lspci` command can be used to verify the device installation status:

```
# lspci -d 1a78:
```

The following output will appear on the screen:

```
82:00.0 FLASH memory: Virident Systems Inc. Virident FlashMAX Drive V2 (rev 01)
```

√ The system should list one device for every device installed.

√ If the device is not detected, power down the server and ensure that it is seated properly in the PCIe slot.

√ Contact *Technical Support* If the device is still not detected.

6.5 FlashMAX II Card(s) Removal

Instructions for removing the FlashMAX II Card(s) can be found in the V2 User Guide.

6.6 Installing Drivers and Utility Packages

There are two `RPMs` that must be installed before the FlashMAX III cards can be used. Ubuntu and Debian Linux Distributions have corresponding `DEB` packages. The driver and utility packages are found on the *HGST Support Portal*.

√ Select the appropriate distribution and kernel version for Linux.

√ Make sure to carefully select the correct kernel version for Oracle Enterprise, Ubuntu, Debian, Fedora Core Linux or openSUSE distribution.

√ The `uname -a` command can be used to confirm the kernel version currently running on the system.

6.7 Additional Information

The file-naming convention is stated below for each of the RPM driver and utility files:

6.7.1 Drivers RPM File-Naming Convention

```
kmod-vgc-drivers-<distribution/kernel version>-<driver version>-<driver build>.x86_64.rpm
```

6.7.2 Utilities RPM File-Naming Convention

```
vgc-utils-< release version>-<release build>.x86_64.rpm
```

6.8 Command Prompt Instructions

Install the driver and utility packages using the `RPM` command for Red Hat or CentOS. Use the `DPKG` command for the Debian or Ubuntu distributions.

6.8.1 Drivers RPM Installation

```
#rpm –ivh kmod-vgc-redhat6.1+-5.2.77498.D1C.x86_64.rpm
```

6.8.2 Utilities RPM Installation

```
#rpm –ivh vgc-utils-5.2.77498.D1C.x86_64.rpm
```

6.8.3 Starting the Driver

The `service` command can be used to start the driver once the packages have been successfully loaded:

```
# service vgcd start
```

The following output will appear on the screen:

```
Loading kernel modules...                     [OK]
Rescanning SW RAID volumes...                 [OK]
Rescanning LVM volumes...                     [OK]
Enabling swap devices...                      [OK]
Rescanning mount points...                    [OK]
```

The drivers will load automatically once the drive and utility packages are correctly installed.

6.9 Restarting the Drivers

The drivers can also be restarted using the `service` command without rebooting the system:

```
# service vgcd start
```

6.10 Listing the Available Devices and Block Devices

The device should be ready for use once the driver has been successfully loaded. The `vgc-monitor` or `ls` commands can be used to list the available devices and block devices.

6.11 vgc-monitor Command

An example of the vgc-monitor command is shown below:

```
# vgc-monitor
```

The following output will appear on the screen:

```
vgc-monitor    : 5.2.77489.D1C
Driver Uptime  : 3:56
  Card Name    Num Partitions      Card Type       Status  Partition  Usable_Capacity    RAID
    VGCA             1         VIR-M3-LP-2200-2B    Good     vgca0         2200GB       Enabled
```

The user will want to confirm that the `Status` of all the listed devices are shown as `Good`.

6.12 `ls` Command

An example of the `ls` command is shown below:

```
# ls -l /dev/vgc??
```

The following output will appear on the screen:

```
brw-rw----. 1 root disk 252, 0 Nov 14 20:35 /dev/vgca0
```

√ While `vgo-monitor` and `ls -l /dev/vgo??` both appear to list the status of all available block devices (physical partitions), each identifies the block device differently.

6.13 Erasing and Upgrading RPMs

There are general rules that should be observed when erasing or upgrading RPMs:

1. Rename the existing config file `/etc/sysconfig/vgcd.conf` as `/etc/sysconfig/vgcd.conf.rpmsave`.
2. As an alternative, save the existing config file to a different location in the event it is needed for any future reference or use. Administrative changes and settings can be preserved through an upgrade or uninstall.
3. The administrator is required to reconcile the new file (installed with the new RPM) and the previous file (the saved rpmsave version from the previously installed version) after an upgrade or new installation.

7. Performance Verification and Validation

7.1 Overview

It is recommended that before a file system is setup on the device that user first measure the raw performance to isolate and resolve any potential issues.

7.2 Test Script

The performance test script, `test.sh`, uses the GPL Licensed FIO Utility and is found on the *Support Portal*. Download and copy the files to the current directory and run it as described in the following sections.

Running the test script will erase or destroy all existing data on the device.

√ ***If the FlashMAX III device is being setup for the first time, then it is recommended to run the test script and then contact Technical Support.***

7.3 Maximum Performance Mode

Maximum Performance Mode provides higher sustained random write performance than that of Maximum Capacity Mode; however, using this mode will reduce the available user capacity of the device by 15% (percent). Maximum Performance Mode should be considered for the following environments:

- While using write intensive applications.
- Workloads that generate higher random write I/Os.
- Higher sustained random write performance.

Be advised that switching between Maximum Capacity Mode and Maximum Performance Mode will erase all existing data on the card.

√ *Read and Sequential Write performance remain constant regardless of the selected mode.*

7.4 Checking the Current Configuration

1. To check the current configuration of all FlashMAX III devices, use the following command:

```
# vgc-config
```

2. The following output will appear on the screen:

```
vgc-config: 5.2.77498.D1C
Current Configuration:
vgca     1 partition(s)
vgca0    mode=maxperformance    sector-size=512     raid=enabled
```

7.5 Specifying the Number of Partitions

The -`n|--num-partitions <num-partitions>` switch in the `vgc-config` command determines whether the device is formatted with one (1) partition or two (2) partitions:

Command Line	Description
`# vgc-config -d vgca0` **`-n 1`** `-m maxcapacity`	Format the device with `one` (**`-n 1`**) partition in `maxcapacity` mode.
`# vgc-config -d vgca0` **`-n 2`** `-m maxcapacity`	Format the device with `two` (**`-n 2`**) partitions in `maxcapacity` mode.

7.6 Specifying the Mode

The `-m|--mode <mode>` switch in the `vgc-config` command will format the partition(s) according to the selected mode. The modes are supported on all partitions of the device:

Command Line	Result
`# vgc-config -d vgca0 -n 1 -m maxperformance`	Formatted for `maxperformance` mode.
`# vgc-config -d vgca0 -n 1 -m maxcapacity`	Formatted for `maxcapacity` mode.

√ *The following sections are used for example purposes only.*

√ *550GB and 1100GB devices only support a single partition of either* `maxperformance` *or* `maxcapacity` *mode.*

7.7 Single Partition Examples

7.7.1 Formatting with a Single Partition in Maximum Performance Mode

1. To format partition `vgca0` on device `vgca` in `maxperformance` mode, use the following syntax:

   ```
   # vgc-config -d vgca0 -n 1 -m maxperformance
   ```

2. The following output will appear on the screen:

   ```
   vgc-config: 5.2.77498.D1C

   *** Formatting drive. Please wait... ***
   ```

7.7.2 Formatting with a Single Partition in Maximum Capacity Mode

1. To format partition `vgca0` on device `vgca` in `maxcapacity` mode, use the following syntax:

   ```
   # vgc-config -d vgca0 -n 1 -m maxcapacity
   ```

2. The following output will appear on the screen:

   ```
   vgc-config: 5.2.77498.D1C

   *** Formatting drive. Please wait... ***
   ```

7.8 Multiple Partition Examples

7.8.1 Formatting Two Partitions in Maximum Performance Mode

1. To format device `vgca` with two partitions in `maxperformance` mode, use the following syntax:

   ```
   # vgc-config -d vgca -n 2 -m maxperformance
   ```

2. The following output will appear on the screen:

   ```
   vgc-config: 5.2.77498.D1C

   *** Formatting drive. Please wait... ***
   ```

7.8.2 Formatting Two Partitions in Maximum Capacity Mode

1. To format device vgca with two partitions in `maxcapacity` mode, use the following syntax:

   ```
   # vgc-config -d vgca -n 2 -m maxcapacity
   ```

2. The following output will appear on the screen:

   ```
   vgc-config: 5.2.77498.D1C

   *** Formatting drive. Please wait... ***
   ```

8. File System Configuration

8.1 Overview

This section explains how to create, mount and modify a file system for a FlashMAX III card. The appropriate number and volumes of partitions are used as either swap space or data space.

8.2 mkfs Utility

The `mkfs` utility is used to create a standard Linux file system on a FlashMAX device.

8.3 Partitions

If only one partition is needed, then it is recommended to create the file system directly on the `/dev/vgca0` or `/dev/vgcb0` device block. Using the `fdisk/gparted` command to create a software partition in such a scenario is neither needed nor recommended.

√ The `fdisk` command does not apply to devices with capacities greater than 2TB; the `parted` command must be used instead.

8.4 parted Utility

The `parted` ***utility manipulates the partition table and immediately saves the changes.***

Incorrect use of the `parted` ***utility can result in data loss.***

Changes made using the `parted` ***utility cannot be undone.***

8.5 XFS: High Performance Mode Enabled

The performance of many applications may be improved if the individual files are aligned on 4KB boundaries. XFS allows for setting the "sector" size equal to 4KB, which guarantees the necessary alignment for all files. It is recommended to use the standard `mkfs` command line for XFS to enable the highest performance. See the following example.

```
# mkfs.xfs -s size=4096 /dev/vgca0
```

√ Database applications that use a direct I/O of more than 4KB size may encounter compatibility issues when using the `–s size=4096` option and might report I/O errors. It is recommended not to use the `–s size=4096` option if such errors are encountered.

8.6 EXT3: High Performance Mode Enabled.

EXT3 can encounter bottlenecks on writes to its journal during high write workloads, especially with high performance devices such as PCIe cards. To help eliminate the bottlenecks, it is recommended that additional space be reserved for the ETX3 journal using the `-J size=xxx` option:

```
# mkfs.ext3 -J size=400 <other standard options as required> /dev/vgca0
```

As an alternative that would be applicable in scenarios where EXT3 journaling is not needed (e.g., scratch areas) is to simply use EXT2 which does not have journaling.

8.7 Auto-Mount Function

To enable auto-mount for a device at system startup, the device entries in `/etc/fstab` must be added or modified.

1. Add the `noauto` entries in `/etc/fstab` with `0 0` options for the FlashMAX III device to ensure that the auto-mount is not attempted before the drivers are loaded during the boot-up process:

   ```
   /dev/vgca0     /mnt1     ext3      noauto, defaults 0 0
   /dev/vgcb0     /mnt2     xfs       noauto, defaults 0 0
   ```

2. Modify the `/etc/sysconfig/vgcd.conf` file to add a list of FlashMAX III devices to a variable known as `MOUNT_POINTS`.

   √ The list of devices should be separated by single spaces.

3. When the driver loads, all the devices in the `/etc/sysconfig/vgcd.conf` file for the variable `MOUNT_POINT` and the corresponding `/etc/fstab` entries will be processed for auto-mount:

   `#` List of mount points that the script should mount `MOUNT_POINT="/mnt1 /mnt2"`

9. RAID Configuration

9.1 Overview

This section explains how to enable or disable the RAID configuration for FlashMAX III devices. The user may want to enable RAID for striping or for mirroring multiple FlashMAX III cards.

9.2 Verifying and Validating Performance

FlashMAX III devices have embedded flash-aware RAID with 7+1P redundancy that provides protection against bad blocks and flash component failures. This will reduce the need for mirroring FlashMAX III devices in certain usage scenarios.

9.3 Creating a Striped Volume

The following example shows how to create a striped volume across two FlashMAX III cards.

```
#mdadm --create /dev/md0 --level=0 --raid-devices=2 /dev/vgca0 /dev/vgca1
```

10. Administrative Utilities

10.1 Overview

A suite of CLI (Command Line Interface) Utilities are delivered with the device drivers to allow users to configure, manage and monitor the FlashMAX III cards.

10.2 vgc-config

The `vgc-config` utility allows the user to switch between Maximum Capacity Mode and Maximum Performance Mode for a device, and to split the physical block device into two partitions of equal size.

Using the `vgc-config` ***utility to change performance modes or manage partitions will result in the erasure of all data on the device.***

√ *The two partitions are presented as separate block devices, i.e.,* `/dev/vgca0` *and* `/dev//vgca1`*; however, in most scenarios, the user is advised to use the default configuration with one physical partition and using software partitions in LVM.*

√ *FlashMAX III devices with capacities of 1100GB and 2222GB do not support the use of multiple partitions.*

10.2.1 vgc-config Help

The `[-h] --help` switch will list the command syntax and options.

```
# vgc-config --help
vgc-config: 5.2.77489.D1C
Usage:
  vgc-config -d|--drive <drive name> [-l|--list]
  vgc-config -d|--drive <drive name>  -r|--reset  [-f|--force]
  vgc-config -d|--drive <drive name>  -b|--boot <enable|disable> [-f|--force]
  vgc-config -d|--drive <drive name>  -m|--mode <mode> [-f|--force] -n|--num-
  partitions <num-partitions>
  vgc-config -p|--partition <partition name> [-l|--list]
  vgc-config -p|--partition <partition name>  -r|--reset [-f|-- force]
  vgc-config -p|--partition <partition name>  -m|--mode <mode> [-f|- -force]
  vgc-config [-h|--help]
Options:
  -d | --drive          : drive name vgc[a,b,---]
  -f | --force          : do not prompt for confirmation
  -h | --help           : display this help and exit
  -l | --list           : list current configuration
  -m | --mode           : maxperformance/maxcapacity mode
  -n | --num-partitions: number of partitions, (1/2)
  -p | --partition      : partition name vgc[a,b,---][0,1,---]
  -r | --reset          : reset to factory defaults
  -b | --boot           : <enable | disable > To set boot flag
No arguments            : list the current configuration of all drives
```

10.2.2 vgc-config Examples

Example: vgc-config used to apply Maximum Performance Mode to the two physical partitions:

```
# vgc-config -d vgca -n 2 -m maxperformance
```

Example: vgc-config used to apply Maximum Capacity Mode to the two physical partitions:

```
# vgc-config -d vgca –n 2 –m maxcapacity
```

10.3 vgc-config controller reset

The `vgc-config controller reset` utility allows the user to revert the device to the factory defaults.

```
vgc-config -r -d vgca
vgc-config: 5.2.77489.D1C

** WARNING: This operation will erase ALL data on this drive, type <yes> to
continue: yes
*** Formatting drive. Please wait... ***
```

10.4 vgc-secure-erase

The `vgc-secure-erase` (Secure Erase) utility will erase (sanitize) all user data on the flash media and will reset the device to its original state in the following scenarios:

- If the device needs to be returned to the factory for repair or replacement.
- If the device is being transferred to another team, project or client for new usage.
- Securely remove all potentially sensitive and proprietary data before being flagged for reuse.

10.4.1 Data Sanitization Standards

Data sanitization standards, as specified by various industrial security, safety and compliance organizations, vary according to the type of media is being used. The standards for magnetic disk media are different than those for flash memory.

The two levels of sanitization used by the FlashMAX III devices are *Clear* and *Purge*. For most cases, the *Clear* level of sanitization will suffice, but *Purge* may be required in other instances.

The specifications as defined by NIST and DOD publications state the following requirements for the two sanitization levels:

Clear: Perform a full chip purge as per the manufacturer's data sheets.

Purge: Overwrite all addressable locations with a single character, followed by Clear.

10.4.2 Government Standards

The `vgc-secure-erase` utility complies with the Clear and Purge levels of sanitization in accordance with the following government standards:

- **DOD 5220.22-M** – Complies with sanitization requirements for Flash EPROM (*http://www.dss.mil/documents/pressroom/isl_2007_01_oct_11_2007_final_agreement.pdf*)
- **NIST SP800-88** – Complies with instructions for Flash EPROM (*http://csrc.nist.gov/publications/nistpubs/800-88/NISTSP800-88_with-errata.pdf*)

10.4.3 Technical Operation

The device should be in a minimal operational state to use the `vgc-secure-erase` utility. The utility will attempt to erase all user data from the device according to the method selected by the user (Clear or Purge).

Notes:

1. The Clear or Purge erase functions may not erase all the blocks due to the following reasons:
   - 1.1. Some blocks may contain user data and are inaccessible due to media failure. *In such a case, an error message will display*.
   - 1.2. Some blocks may have been marked as unusable at the time of shipment from the factory; the system cannot write to these bad blocks. *In such a case, an error message will not display*.
2. In either of the aforementioned cases, the blocks cannot be 'sanitized' but it is guaranteed that such faulty blocks never received any user data.
3. The `vgc-secure-erase` (Secure Erase) utility can still sanitize a device when the blocks are present.

10.4.4 Command Help

Use the `--help` switch to list the online help for the `vgc-secure-erase` utility:

```
# vgc-secure-erase --help
vgc-secure-erase: 5.2.77489.D1C

Usage:
  vgc-secure-erase [-c|--clear] [-y|--yes] <partition>
  vgc-secure-erase [-p|--purge] [-y|--yes] <partition>
  vgc-secure-erase [-h|--help ]

Options:
```

`- C | -- clear:`	Clear (**Erase**) the contents of the partition. ***This is the default option***.
`- H | -- Help:`	Displays options/parameters that vgc-secure-erase can take.
`- P | -- Purge:`	Purge (**Erase & Overwrite**) the contents of the partition.
`- Y | -- Yes:`	Start the operation without user confirmation.
`- V | -- verify:`	Read and verify after erase operation done.
`partition:`	Name representing the partition instance (e.g. vgca0).

Notes:

1. If no option is given, the 'Clear' operation is performed on the given partition by default.
2. The operation physically erases all data from the given partition. It physically erases all data from the given partition. This operation may take several minutes.
3. Run `vgc-secure erase` on all partitions to securely erase all data.

Be advised that the operation is destructive. The data cannot be recovered after the Clear or Purge operation.

Figure 5 shows the results of a secure erase operation.

```
# vgc-secure-erase vgca0
vgc-secure-erase: 5.2.77489.D1C

This operation will erase all data on the physical partition vgca0.
It will take up to 20 minutes to complete.
Once started you cannot stop or undo this operation.
Do you want to continue? [yes/no] Yes
[|-|]
Starting Secure Erase (Clear) operation Please wait...
Result:
Secure Erase (Clear) on partition vgca0 completed successfully.

Operation Summary:
Number of erase operations skipped due to factory bad blocks: 435
Number of erase operations skipped due to grown bad blocks  : 0
Number of erase operation failures                          : 0
Reformatting the partition vgca0

#vgc-secure-erase vgca0 --yes
vgc-secure-erase: 5.2.77489.D1C
[|-|]
Starting Secure Erase (Clear) operation. Please wait...
Result:
Secure Erase (Clear) on partition vgca0 completed successfully.

Operation Summary:
Number of erase operations skipped due to factory bad blocks : 435
Number of erase operations skipped due to grown bad blocks   : 0
Number of erase operation failures                           : 0
Reformatting the partition vgca0

Starting Secure Erase (Clear) operation. Please wait …
Result:
Secure Erase (Clear) on device vgca0 completed successfully.
Reformatting the device vgca0
```

Figure 5: Secure Erase Operation

10.4.5 Command Failure

The `vgc-secure-erase` command usually fails because the device is still mounted:

Command Output:

`/dev/vgca0: Device or resource busy. Secure Erase (Clear) on device /dev/vgca0 failed.`

Check kernel syslog for more details using the following command:

```
# syslogd [-a socket ] [-d] [-fconfig file] [-h] [-l hostlist] [-m interval] [-n] [-p
socket] [-r] [-s domainlist] [-v] [-x]
```

Unmount the device before secure erase using the following command:

```
# umount/dev/vgca/mnt
```

10.5 vgc-monitor

The `vgc-monitor` utility allows for the monitoring of device status and health. If the command is used without any parameters, it will list the statistics for all FlashMAX III devices in the system:

```
#vgc-monitor --help
vgc-monitor: 5.2.77489.D1C

Utility to monitor FlashMAX III drives

Usage:
  vgc-monitor [-d|--drive <drive-name>][-f json | tty]
  vgc-monitor [-h|--help]

Options:
 -d |--drive: drive name vgc[a,b,---] display info for the given drive
 -h |--help: display this help and exit
 no arguments: display info for all drives
```

Figure 6 shows an example of the command output.

```
Device specific:
# vgc-monitor -d vgca
vgc-monitor              : 5.2.77489.D1C
Driver Uptime            : 4:03

Card_Name                  Num_Partitions         Card_Type        Status
/dev/vgca                        1            VIR-M3-LP-2200-2A    Good

Serial Number            : SMT00179
Card Info                : Part: HIT50-02106-040U
                           Rev: FlashMAX II 74847, x8 Gen2
                           Boot ROM: not present
Temperature              : 49 C (Safe)
Temp Throttle            : Inactive
Card State Details       : Normal
Action Required          : None

Partition                Usable Capacity     RAID
vgca0                    2222 GB             Enabled

Mode                     : maxcapacity
Total Flash Bytes        : 547496081146880 (547.50TB) (reads)
                           759364463001600 (759.36TB) (writes)
Remaining Life           97.92%
Partition State          READY
Flash Reserves Left      99.93%
```

Figure 6: vgc-monitor Command Output

10.6 vgc-diags

The `vgc-diags` utility will compile a report of support data, i.e., log files, system and device configuration, and device metadata into a single file named `vgc.diags.tar`. If necessary, the report file can then be sent to technical support for analysis.

```
# vgc-diags --help
vgc-diags: 5.2.77489.D1C

Usage:
vgc-diags -p|--outputpath <dump directory> [-v|--verbose] [device list]
vgc-diags [-h|--help]

Options:
-h| --help        : Display this help and exit
-v| --verbose     : Verbose, dumps detailed information
-p| --outputpath: Directory to save the diags file device
List              : List of devices in the format vgc[a,b, ---]
                  : Default is all FlashMAX III devices detected in the system
```

Notes:

1. The `–v` option may result in the report file size that is several gigabytes in size.
2. The `–v` option should not be used unless instructed otherwise by a support engineer.

10.6.1 Saving Diagnostic Information for all Devices

To save the diagnostic information for all FlashMAX III devices installed in the system:

```
# vgc-diags
```

10.6.2 Saving Diagnostic Information for a Single Device

To save the diagnostic information for a single device, specify the device name after the command:

```
# vgc-diags vgca
```

10.7 vgc-beacon

The `vgc-beacon` utility facilitates multiple-device installation and will identify a single device by manually flashing the on-board LEDs.

Notes:

1. Execute the `vgc-beacon` command with the desired "`vgc[a, b, ---]`" device and "`–b 1`"or "`–b 0`" to enable or disable the beacon respectively.
2. When enabled, two on-board LEDs will flash in parallel to identify the FlashMAX III card.
3. If "`–b 1`" or "`–b 0`" is not used the command will return its current settings.

10.7.1 Example: Activating the Beacon

For example, to activate the beacon for device `vgca`:

```
# vgc-beacon vgca -b 1
```

10.7.2 Example: Deactivating the Beacon

For example, to deactivate the beacon for device `vgca`:

```
# vgc-beacon vgca -b 0
```

10.7.3 vgc-beacon Help

```
# vgc-beacon --help
vgc-beacon: 5.2.77489.D1C

Usage:
vgc-beacon-d|--drive <DriveName | PCI Address> [ -b|--beacon 1|0>(enable/disable
beacon) ]

Options:
-b| --beacon: <1|0>, Enable/Disable beacon. (1 -> Enable, 0 -disable.
-d| --drive : Drive name(vgc[a,b,---])or PCI Address{domain:bus:dev.fn}
-s| --show  : Show Hardware Id of all cards with their respective PCI Address.
-h| --help  : Display this help and exit.
```

11. Dynamic Throttling Management

11.1 Overview

This section explains how to manage power and thermal throttling for the FlashMAX III.

√ *The default values for the throttling levels are set at the factory.*

11.2 Dynamic Throttling

Dynamic Throttling will prevent unplanned or abrupt device shutdowns by lowering the IO performance should the power consumption and/or the temperature increase. Dynamic Throttling is comprised of thermal and power throttling.

11.2.1 Thermal Throttling

Thermal Throttling will reduce the IO performance of the device if the temperature exceeds the default threshold of 78°C.

√ *Once the device has cooled and the operating temperature has fallen below the default T[throttling] level, the IO performance will rise to its maximum possible level (100%) in incremental steps of 10% every 900 seconds.*

√ *A card that has undergone a shutdown due to reaching 85°C requires manual intervention before it will come online.*

- If the temperature continues to increase and reaches 83°C, then the device is throttled down to the lowest IO performance.
- If the temperature reaches 85°C, then the device completely shuts down.
- The `service vgcd start` command is required to restart a device that has undergone a complete shutdown.

11.2.2 Preventing Thermal Throttling

There are several approaches to prevent thermal throttling:

1. Choose a slot for the card where the airflow for the heat sink is not obstructed.
2. If possible, the ambient air temperature of the environment should be reduced to prevent thermal throttling.
3. If possible, extra fans or additional ventilation should be added to the server chassis to enhance the air circulation around the devices.
4. If the design of the server chassis permits, the cards can be re-arranged in order of size to allow for suitable air flow about the devices.

11.2.3 Power Throttling

Power Throttling will reduce the IO performance of the device should the power consumption exceed the default threshold of 24W. The FlashMAX III has two modes: Turbo Mode and Compliant Mode:

Turbo Mode: Throttling is disabled.

Compliant Mode: The IO level of the device is reduced to the highest level while maintaining the power level threshold.

√ *Once the power consumption is less than or equal to the threshold value, the IO performance returns to its maximum possible level in incremental steps of 1% every 120 seconds.*

√ *The default values for the throttling levels, such as T[throttle], T[offline], and the Turbo/Compliant Modes, are set at the factory.*

12. Appendix A: Troubleshooting

12.1 Overview

This section explains the general troubleshooting methodology the user can employ to diagnose problems with the FlashMAX III device. The main issue a user may encounter is that the device or driver does not appear to be functioning properly or is not recognized by the system. There may also be performance issues with the device. The user should be familiar with the basic Linux and FlashMAX III administrative utilities to resolve any issues.

√ *If the user cannot resolve the issue after following the instructions in this section, then it may be necessary to contact Technical Support.*

√ *The technical support staff may require the user to collect all the relevant system information using the* `vgc-diags` *utility to conduct a failure analysis.*

12.2 Checking the Hardware Installation

If the device is not found or recognized by the system, then follow these steps:

1. Power down the system and make sure that the device is seated properly in the PCIe slot.
2. If the device is seated correctly in the slot, but still cannot be detected by the `lspci` command, then try a different slot.
3. If the device is still not detected after changing the slot, then contact Technical Support.

12.3 Using the lspci Command

The `lspci` command can be used to determine the installation status of PCIe devices:

```
# lspci -d 1a78:
```

The following command output should display on the screen:

```
82.00.0 FLASH memory: Virident Systems Inc. FlashMAX Drive V2 (rev 01) (rev 01)
```

12.4 Checking the Driver and Utility RPMs Installation

If the device was detected by the lspci command, check to determine that the correct driver and utility RPMs are installed.

√ *The device driver will not load if the kernel version is not compatible with the FlashMAX III software.*

The `rpm` command can be used verify the installation of the correct driver and utilities:

```
# rpm -aq | grep vgc
```

The following command output should appear on the screen.

```
# rpm -aq | grep vgc
vgc-utils-5.2-77489.D1C.x86_64
kmod-vgc-2.6.32-431-e16.6.1.x86_64-5.2-77489.D1C.x86_64
```

12.5 Checking the vgcd Service

1. Verify that the `vgcd` service is running using the following command:

   ```
   # service vgc -d status
   ```

   The following command output should appear on the screen:

   ```
   INFO: kernel modules are loaded.
   ```

2. If the kernel modules are not loaded, then manually start the service:

   ```
   # service vgcd start
   ```

   The following output is displayed on the screen:

   ```
   Loading kernel modules...                                 [OK]
   Rescanning SW RAID volumes...                             [OK]
   Rescanning LVM volumes...                                 [OK]
   Enabling swap devices...                                  [OK]
   Rescanning mount points...                                [OK]
   ```

12.6 Checking the Syslog for VGC Error Messages

If manually starting the service results in a failure, then check `syslog` for VGC error messages:

```
# cat /var/log/messages | grep -i vgc
```

If any of the following messages are listed, then try the following solution(s).

1. "INFO. Drive 'a' (PCI 0000:82:00.0 rev 01) controller 0. Installed firmware is compatible but recommended firmware version is: 60174, 52879 and 479."

   Note: Firmware mismatch message(s) will display for other firmware versions.

   Solution: Update the controller firmware.

2. "vgcd: Not loading kernel modules because the previous load was unsuccessful. Check the system logs for details. To force load, remove `/var/lib/vgc/loading.lck`"

   Note: The user should contact Technical Support if unsure how to create a `lck` file.

   Solution: Remove the `loading.lck` file and issue a `service vgcd start`.

3. "vgcd: FATAL: Module vgc drive not found."

   Note: Review the instructions as per *Installing Drivers and Utility Packages*.

   Solution: Uninstall the `rpms` (driver and utilities) using the following command line and then re-install the `rpms`:

   ```
   rpm -aq | grep vgc | xargs rpm –ev
   ```

12.7 Checking the Device Health

The `vgc-monitor` utility can be used the check the health status of the device.

```
# vgc-monitor -d vgc[a,b,c---]
```

√ *If the device status is "*`Critical`*" or "*`Warning`*", then check the "*`Action Required`*" field in the command output of the* `vgc-monitor` *utility for instructions.*

12.8 Checking the Device Performance

If the initial *Test Script*, `test.sh`, reports low system performance, or if the installed application is not performing as per expected benchmarks, then verify that the device is installed in a PCIe slot which is x8 electrical.

There are certain system motherboards that provide x8 slots that are physically the correct size, but only connect x4 PCIe Lanes. The user should also check the BIOS settings to ensure that the maximum performance option(s) are enabled. *See Selecting a PCIe Slot and Configuring System BIOS Settings*.

The system manual or the motherboard can provide the best reference. The electrical connections are often labeled next to the slot(s) on the motherboard.

√ *The choice of file system can also affect the device performance. See File System Configuration.*

12.9 Driver Crashes upon Booting

If this condition occurs when the driver attempts to load upon the system boot, then run the following command to ensure that the driver does not load automatically:

```
# chkconfig vgcd off
```

Set the device driver to load automatically after the system has been debugged and the issue is fixed:

```
# chkconfig vgcd on
```

12.10 Driver Hang Stops IO Processing

If the IO processing stops because the driver is hung, a message similar to the following will be logged:

```
kernel: INFO: task vgc_mtwa0:13866 blocked for more than 120 seconds.
```

1. Run the `vgc-diags` utility to collect debugging information for the Technical Support personnel and then reboot the system:

   ```
   # vgc-diags
   ```

   ```
   vgc-diags: 5.2.77489.D1C
   Creating vgc.diags.tar
   Collecting diagnostic data...
   ```

2. Send the compiled `vgc.diags.tar` file to Technical Support.

12.11 Other Errors

For all other errors, check the output of the `vgc-monitor -d vgc[a-x]` utility and note the "Action Required" for the steps needed to resolve the issue.

13. Appendix B: Diagnostic LEDs

13.1 Overview

There are two on-board LEDs on the FlashMAX III device.

- The Amber (WRITE LED) is on the top side, while the Green (READ LED) in the middle.
- These LEDs can also be used to identify individual devices in a multi-device installation using the `vgc-beacon` utility.
- The LEDs in FlashMAX III are reversed from that of FlashMAX II.

13.2 LED Indicator Annunciation

Table 4 explains what each combination of LEDs indicates.

Table 4: LED Indicators

LED Color		Status Indicated
Green	**Amber**	
OFF	OFF	No power to the device.
ON	ON	Power on, no driver loaded.
ON	OFF	Power on, driver loaded.
FLASH	-	Data being read.
-	FLASH	Data being written.
SLOW FLASH	SLOW FLASH	Beacon ON.

14. Appendix C: Quality Certifications

14.1 Overview

This section describes the marks and approvals that may be found on the FlashMAX III device.

- FlashMAX III WHQL Certified
- FlashMAX III is UL Certified

14.2 Certifications

The following certification marks usually appear on the device:

14.3 Technical and Compliance Standards

Table 5 lists the technical and compliance standards of the device.

Table 5: Technical and Compliance Standards

Country	Applicable Quality / Compliance Standard
USA/Canada	UL 60950-1, 2nd Edition & CSA C22.2,
	EN60950-1:2006 + A1:2010 + A11:2009 + A12:2011 cULus
	FCC Part 15 Subpart B Section 15.107/109/ANSI C63.4 (2009), Class B
	ICES-003 Issue 5, Class B
	EN 55022,Radiated & Conducted Emissions Class B
	EN 55024, Radiated Immunity & ESD
Europe	CAN/CSA-C22.2 No. 60950-1/A1:2011
	EN60950-1/A12:2011
	cTUVus
Japan	VCCI - V-3/2012.04 Class B
Taiwan	BSMI CNS 13438
Korea	KCC – KN 22/KN61000 – KCC-REM-ST6-PCIe
Australia/NZ	ACMA – EN55022:2010 and EN55024:2010 (ESD and Radiated Immunity)
	RoHS Directive 2011/65/EU
	REACH EU Directive EC No.1907/2006
	WEEE (DIRECTIVE 2012/19/EU)

15. Contact Information

15.1 General Information

Main Web Site: *http://www.hgst.com*

15.2 Technical Support

Main Support: *http://www.hgst.com/support*

Technical Support: *http://www.hgst.com/support/contact-support/tech-support*

15.3 Email Support and Telephone Support

Region	Email	Telephone
North America	*support_usa@hgst.com*	**Toll Free**: +1 888 426-5214 **|** **Direct**: +1 408 717-8087
Asia Pacific	*support_ap@hgst.com*	+65 6840 9595
EMEA / UK	*support_uk@hgst.com*	+44 20 7133 0032
Germany	*support_de@hgst.com*	+49 6929 993601

Index

HGST, a Western Digital company
3403 Yerba Buena Road
San Jose, CA 95135
Produced in the United States

United States: 800-801-4618 (Toll-Free)
International: 408-717-6000

22 December 2014